

Business Management Systems

Networks

Riaz Esmailzadeh

We respectfully acknowledge the Kaurna, Boandik, and Barngarla First Nations Peoples and their Elders past and present, who are the Traditional Owners of the lands that are home to our campuses across Adelaide and South Australia.

Outline

- Product costs – fixed and variable
- Breakeven points
- Some economics
- Price discrimination
- Auctions

Fixed and Variable Costs

- Penguin Random House paid Michelle Obama some \$30 million to write her memoir in the White House (not sure about these figures, but let's assume.)
- They further promoted the book to a global audience, costing another \$20 million.
- Each book sold for \$50, of which Obama received 50% royalties
- Retailers usually keep 30% of the sales.
- How many copies should Penguin sell to break even?



Fixed Costs

- Fixed costs, or sunk costs, or overhead cost are those incurred irrespective of the number of product units
 - Pre-payment to Michelle Obama has no relation to the number of books that gets sold
 - Examples?

Variable Costs

- Variable Cost on the other hand are the cost directly associated with a unit of the product being sold, such as
 - Cost of production, including cost of raw material
 - Percentages paid for patents, royalties, etc
 - Percentages paid to retailers, wholesalers, etc.

Breakeven Point

- The presence of fixed cost in making and delivering any product means that there always exist a breakeven point
- That is, you must sell a minimum number of units to break-even: to re-coup the costs of making the product
- Any unit that you sell thereafter will then directly contribute to profits
 - The minimum number of required sales determines whether you will or will not release a product to a market: what is the point in making something that needs to sell a 1000 units to break-even, to a population of 500 (or to a target market of 500).

Breakeven Calculation

- You can calculate the number of units you need to sell to break even from:

$$\text{Breakeven Units} = \frac{\text{Fixed Cost}}{\text{Unit Price} - \text{Variable Cost}}$$

- Any unit that you sell thereafter will then directly contribute to profits

Fixed Cost Business

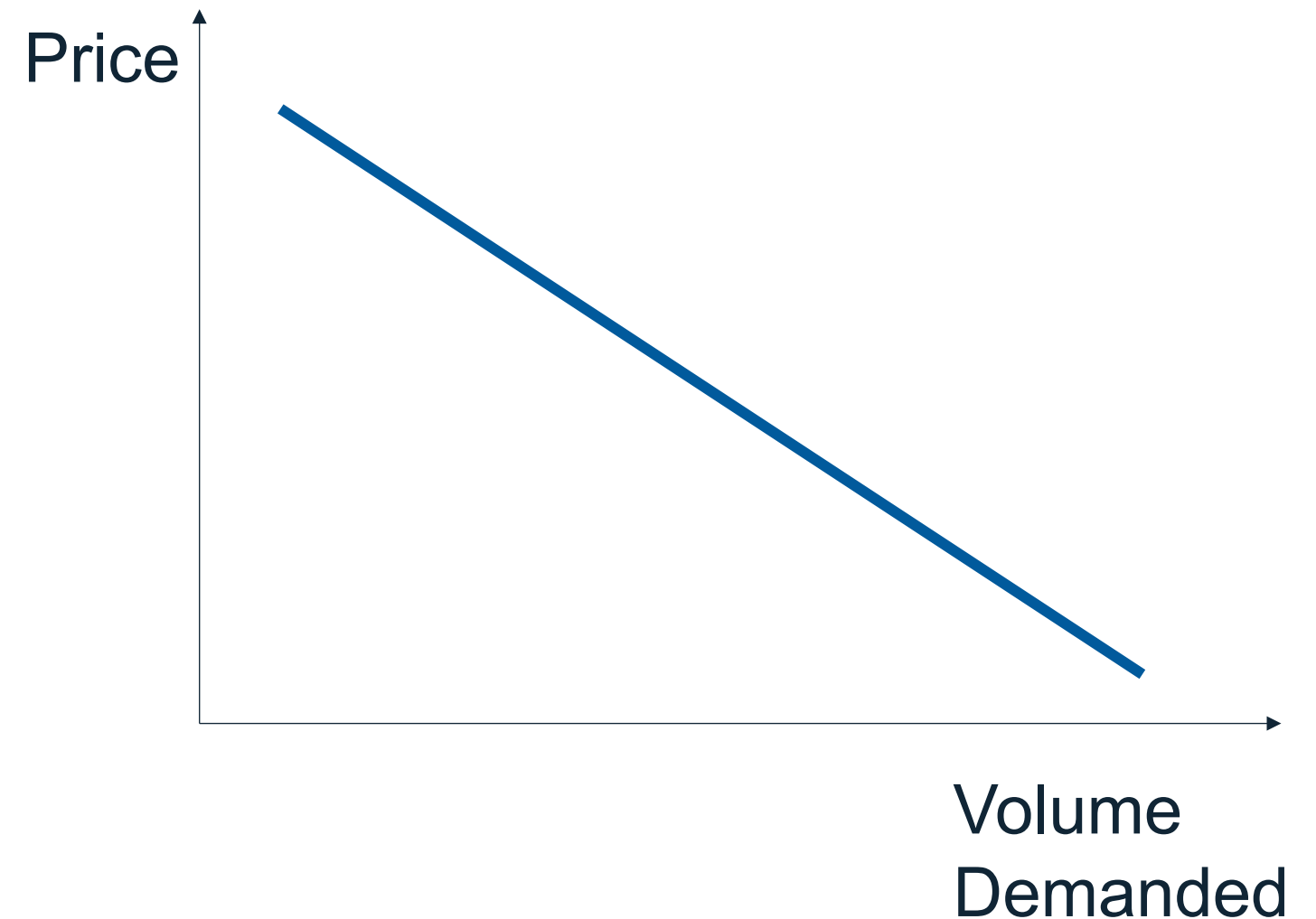
- A fixed cost business is one where the cost of the products it sells are primarily fixed.
- Can you think of an example?

Variable Cost Business

- A variable cost business is one where the cost of the products it sells are primarily variable.
- Can you think of an example?
- In general, most products have a variable and fixed cost component. However, some have one dominant component

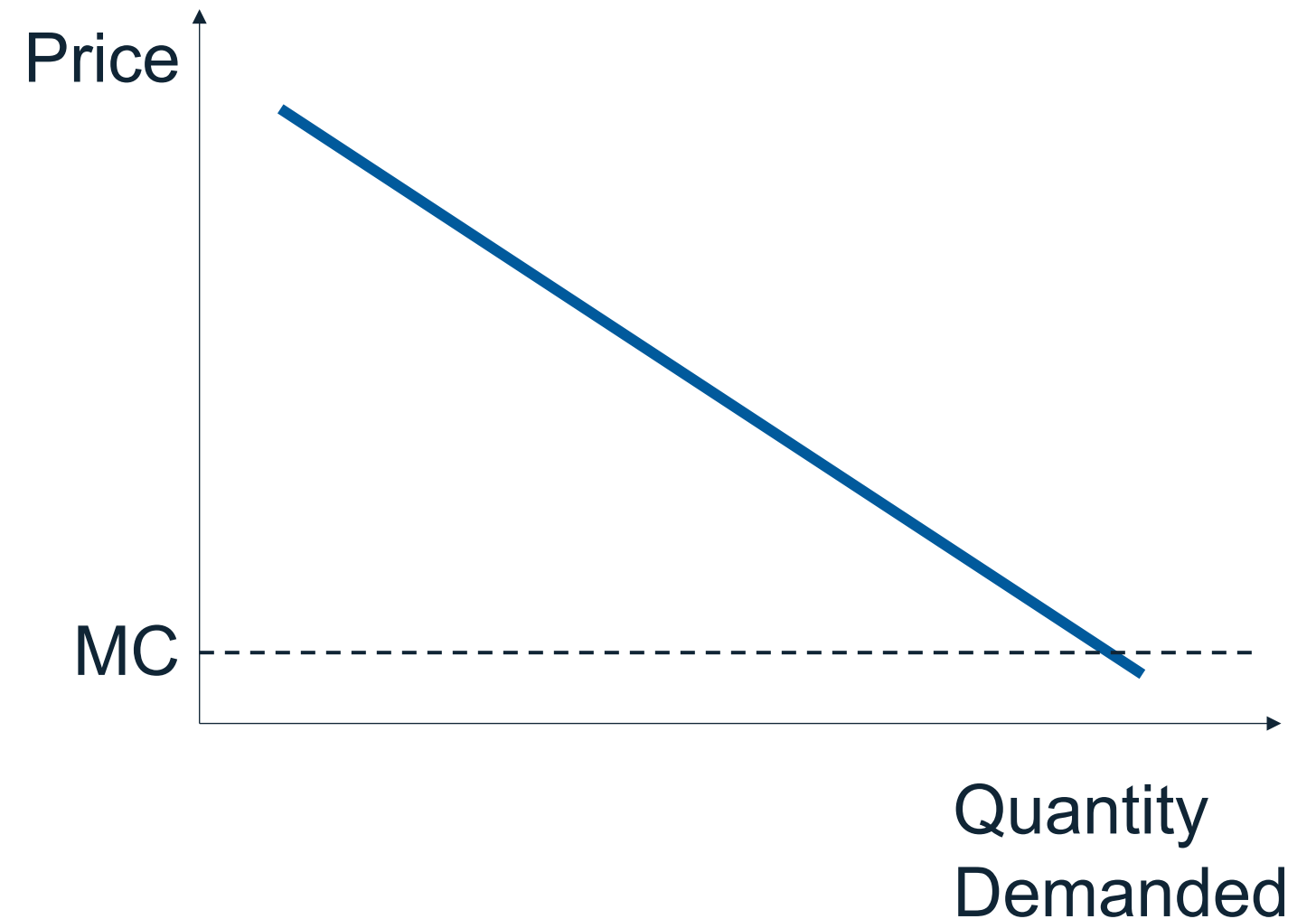
Demand Curve

- Demand curve represents a fundamental concept in Economics
- As the price of a good/service increases, demand for it decreases
- Willingness to Pay (WTP) price determines whether one buys a good/service or not



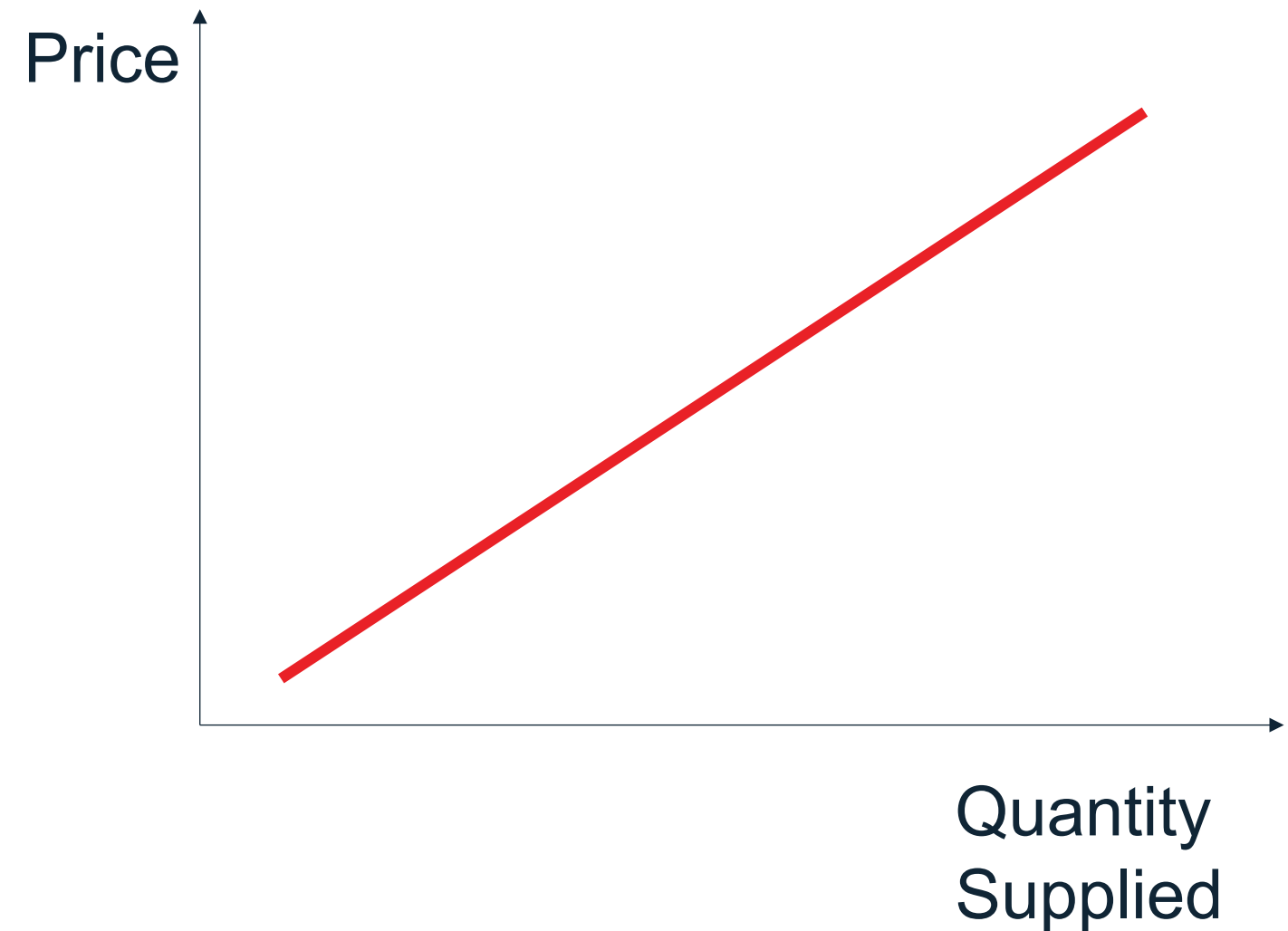
Marginal Cost

- Clearly the price cannot decrease to zero
 - Not accounting for services that are subsidized otherwise... (examples?!)
- There exist a marginal cost (MC) floor, below which the product cannot be offered as a loss



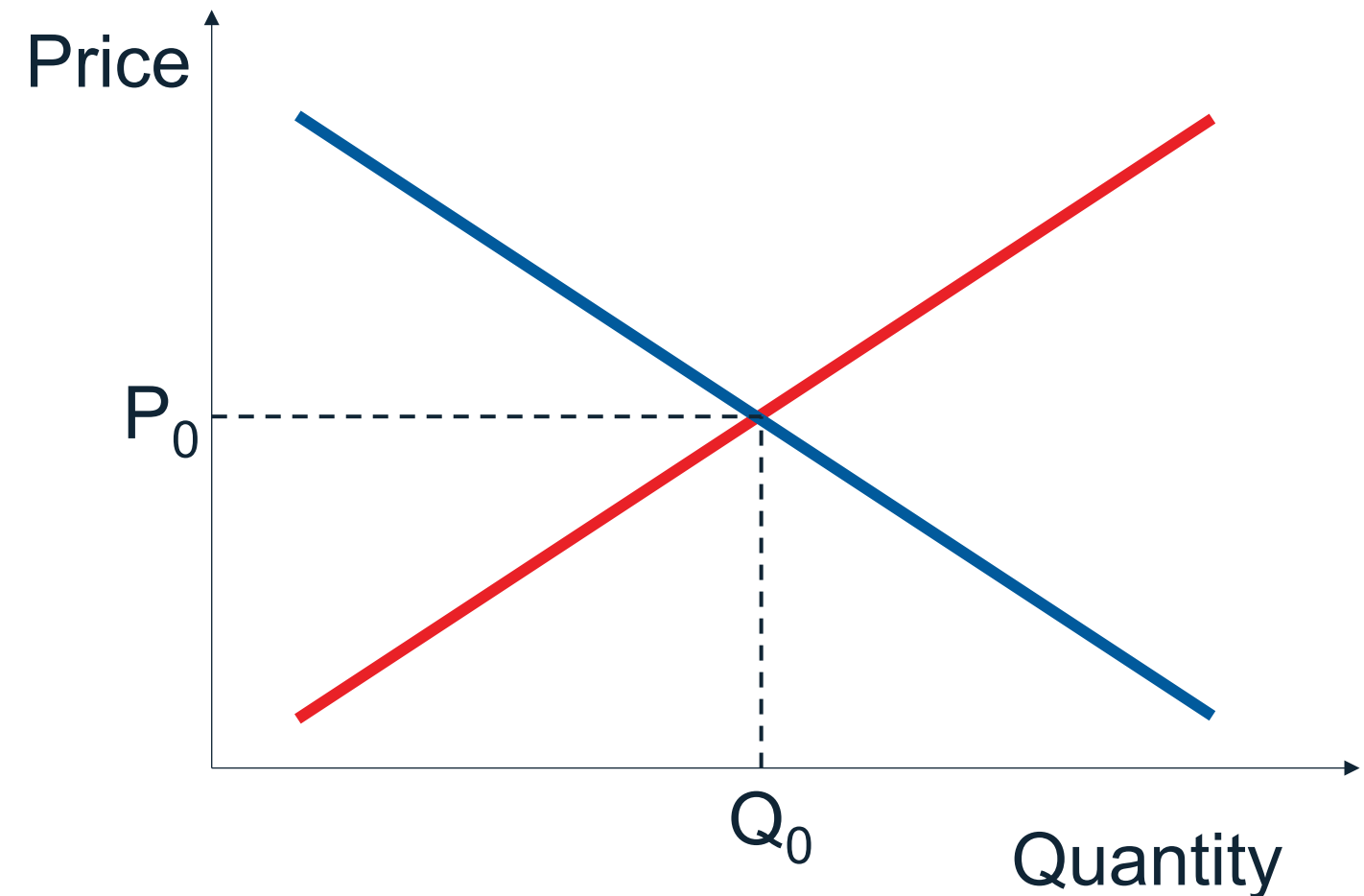
Supply Curve

- Supply curve represents another fundamental concept in Economics
- As the price of a good/service increases, (more) supplier(s) produce more of it, and supply it to the market
- Here also other factors come into play, as resources are always limited



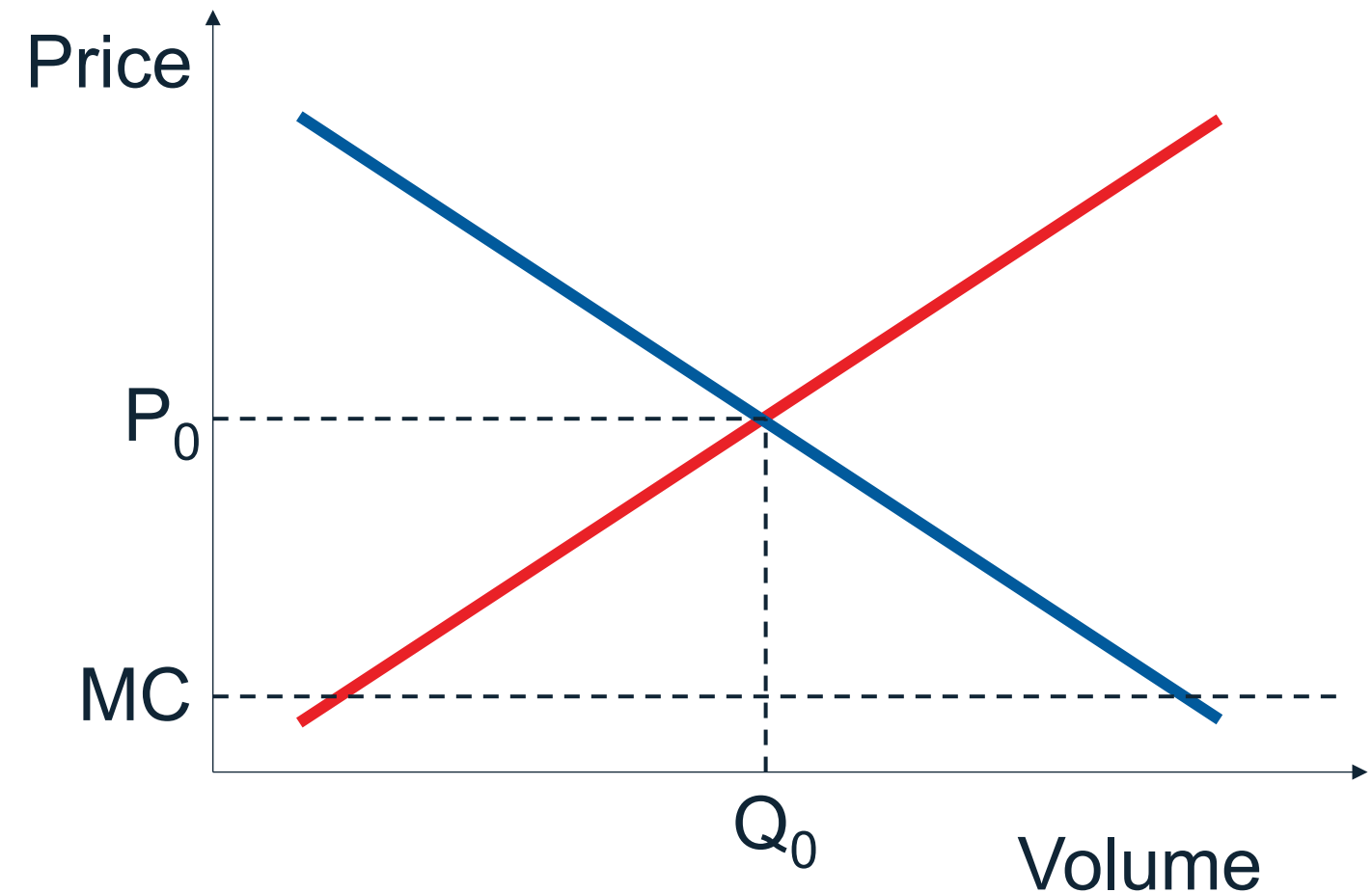
Equilibrium Price

- The price point where equal units of a good are demanded and supplied is known as the equilibrium price (P_0)
- Market forces work to balance needs and interests of consumers and producers



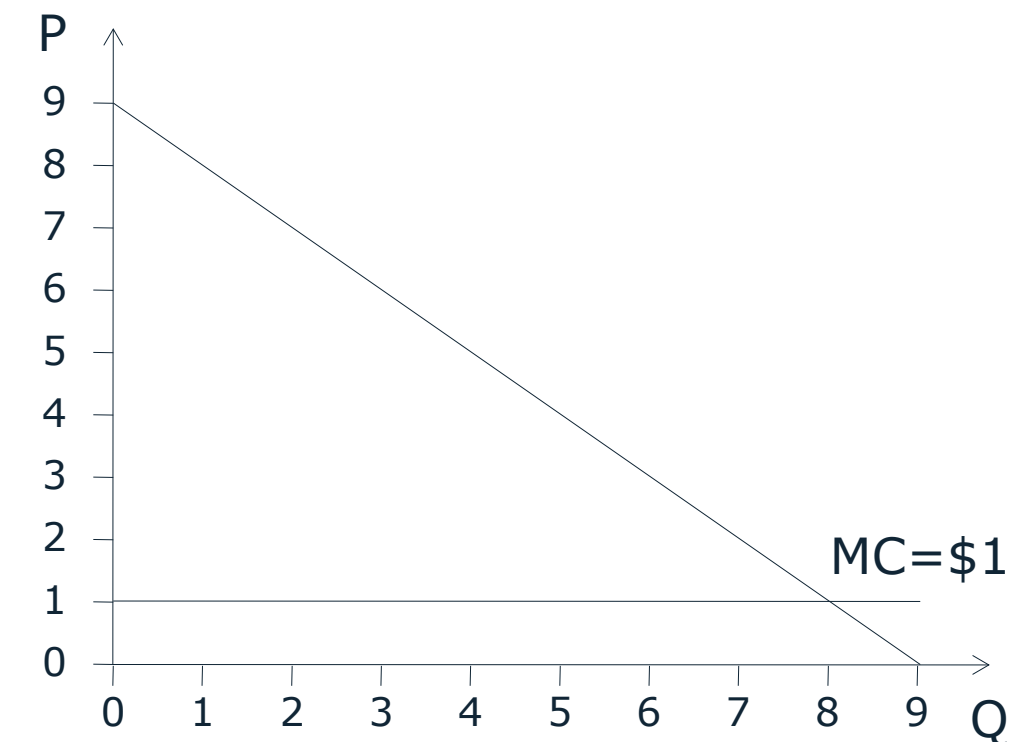
Unfulfilled Demand

- Note there is a price point area where producer are willing to supply goods/services, and consumers are willing to pay.
- However, the equilibrium price does not allow this segment to be served
 - What do we do?!



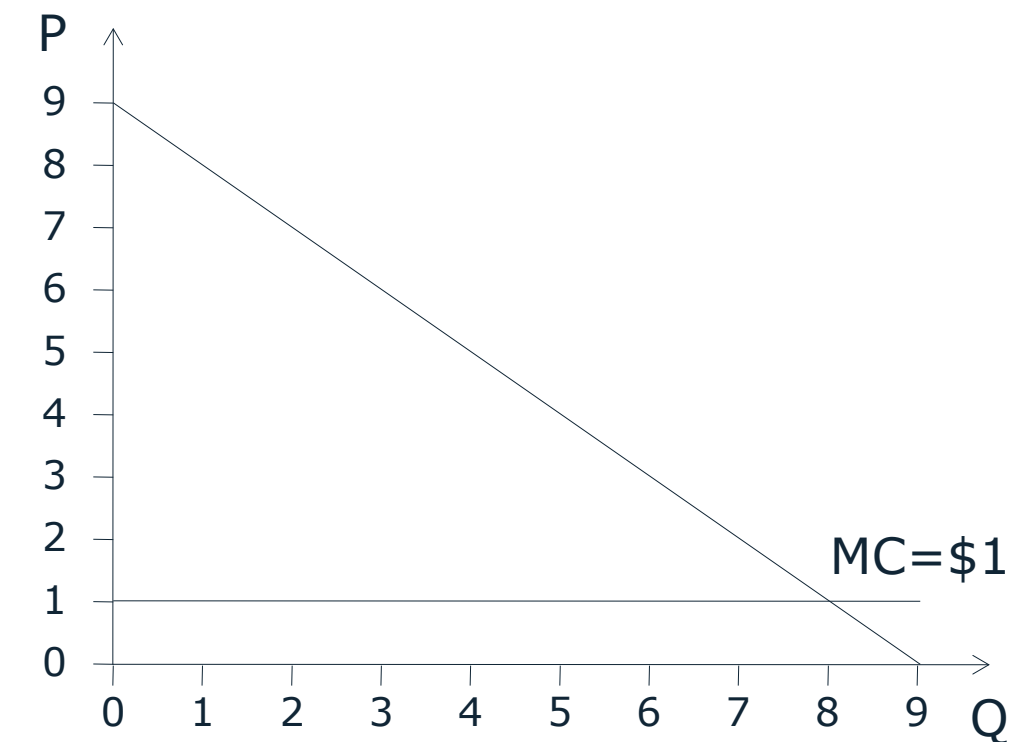
Example

- Let the demand for a product be as shown.
Marginal cost (MC) for producing a unit is \$1.
- Let there be customers with WTP's of 8, 7, 6, etc.
- The equilibrium price that maximizes profits = \$5
 - Profit at that price is \$16.
- However, if the producer charged a different price to each customer. Say \$7 to the customer with WTP of \$8, \$6 to a customer with WTP of \$7 and so on till the buyer with a WTP of \$2. Profit is now $7 + 6 + 5 + 4 + 3 + 2 + 1 = \28



If Only We Could!

- Any seller would love to have a mechanism so it can extract maximum willingness to pay price from its customers
- Price discrimination tries to do just that: selling different units of goods at different prices to different customers



Types of Price Discrimination

- Three types of price discrimination is defined:
 - First degree price discrimination: sell to each customer at their WTP price level.
 - Difficult since people usually do not reveal their WTP
 - Second degree price discrimination: create different category of prices for the 'same' product, and let them select which they would buy
 - Not very difficult: the trick is to create sufficiently differentiated set of product
 - Third degree price discrimination: create different segments of customers and set different prices for each segment
 - Not very difficult: the trick is creation of legitimate discrimination categories

First-degree Price Discrimination

- If only we could somehow find a buyer's WTP...
 - Well we can: through
 - Negotiation
 - Auctions
 - Tracking buyer's behaviour (Amazon, Facebook, Google)
 - Usually for 'unique' goods
 - Value-based priced



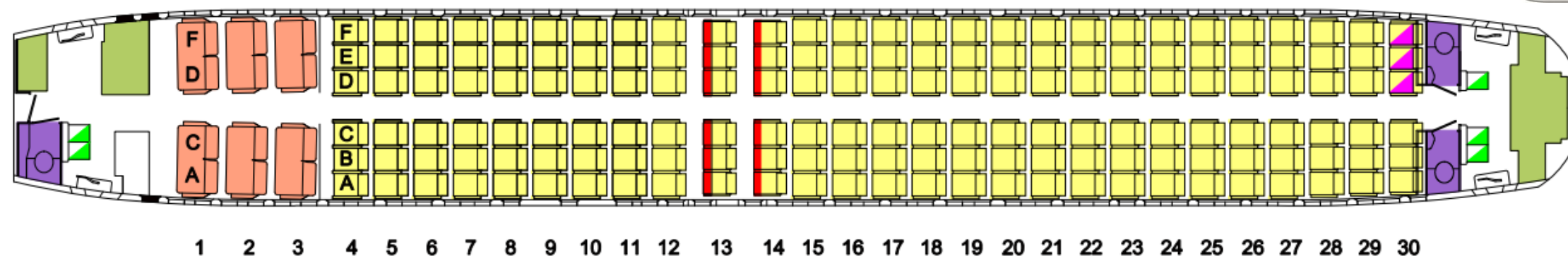
Second-degree Price Discrimination:

- If we know overall market segmentation, we can produce goods which are sufficiently differentiated, and set different prices for them
 - Different sized storage, size, etc
 - Different class of travel, seat row, etc
 - Both are high fixed price businesses



737-800

■ 12 BUSINESS
■ 162 ECONOMY



Third-degree Price Discrimination

- When different legitimate segments exist, a different price may be offered to each group
- At times, different group of people may be charged a different price
 - For example: tourists vs locals



Deadpool & Wolverine

Showing on Sun 18 Aug 6:45PM
Mitcham - Cinema 1

SELECT TICKETS

Select the number and type of tickets you wish to buy. Please note seats are reserved on a best available basis. You can buy a maximum of 10 tickets per transaction.

Please enter your [VOUCHER BARCODE](#) or your [CODEWORD](#) in the Voucher Tab below.

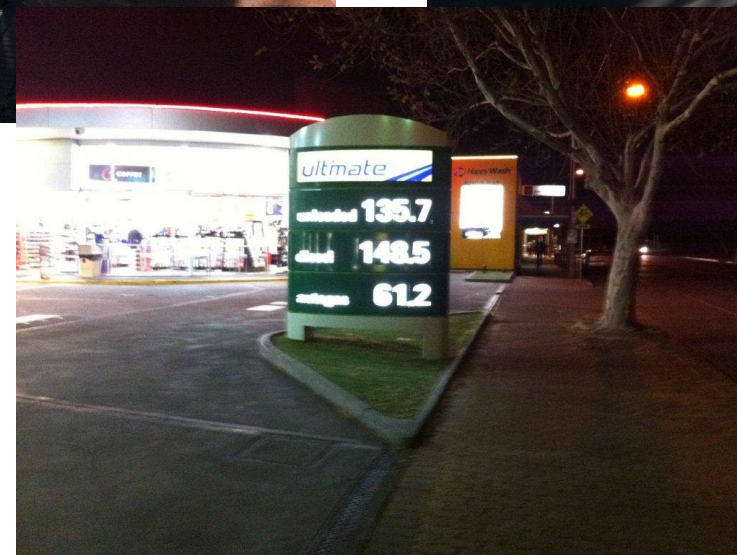
Wallis Loyalty

You are currently not signed in. [Sign In](#)

Standard	Tickets	Cost	Qty	Subtotal
Voucher	Adult Ticket	20.00	− 0 +	0.00
	Student/Concession	18.00	− 0 +	0.00
	Child Ticket	16.00	− 0 +	0.00
	Senior Ticket	15.00	− 0 +	0.00

What Type of Price Discrimination Is This?!

- These photos are a few years old: I took them as I walked from/to home/work on five consecutive days – note petrol is a variable price product



What Type of Price Discrimination Is This?!

- Another example of a variable price product



EVERYDAY LOW PRICE

\$3.90
\$1.95 / 1L

Coca - Cola Classic Soft
Drink Bottle 2l



\$3.85
\$3.08 / 1L

Coca - Cola Classic Soft
Drink Bottle 1.25l



\$4.25
\$7.08 / 1L

Coca - Cola Classic Soft
Drink Bottle 600ml



\$9.15
\$6.93 / 1L

Coca - Cola Classic Soft
Drink Multipack Glass
Bottles 4 X 330ml



Special

SAVE \$7.60

\$11.40
\$19.00 - \$3.17 / 1L

Coca - Cola Classic Soft
Drink Mini Bottles 300ml X
12 Pack

Auctions

- Auctions are the most common way of finding a customer's WTP
- There exist many different types
 - Open vs. sealed bid
 - Ascending price vs. descending price
 - Single-round vs. multi-round
 - Single item vs. multiple items – used in selling bonds
 - Sealed bid second-price (also known as Vickrey) – used by Google

The English Auction

- In an open cry (English) auction the seller start by offering a good for sale
 - Commonly used for arts, real-estate, (eBay)
- The final sale price is determined through a bid process
- While the offer is on buyers can continue to raise the price
- The process ends by when no-one raises the bid, at which time
 - The seller accepts: a sale
 - The seller withdraws offer: no sale

The Dutch Auction

- Auctioneer starts with a certain price, say \$1000
- Price lowers gradually (auctioneer counts back from \$1000) until some buyer signals mine
- The buyer to signal gets the object at the price the auctioneer called just before the signal was made
- This reduces the likelihood the price is below one's WTP

Multi-Round Sealed Auction

- In this process, the auction follows a similar way to the English Auction as before, expect that at regular intervals, the seller publishes the latest bids
- An example is the auction for frequency spectrum in Australia
- Spectrum in each major area is offered for sale
- As long as there are bidding going for any area, no final sale is made
- When highest prices are decided, then entire spectrum auction is finalised.

Dutch Multiple Item Auctions

- When there are multiple number of items, an auction process may allocate the good to multiple buyers
- Example: I have 500 chairs to sell!
 - Rules: minimum price \$25, minimum order 40, maximum order 100 units

Dutch Auction: Sale of Bonds

- Governments and companies around the world raise funds through selling of bonds using Dutch Auction.
- For example, suppose the Australian government is seeking to raise \$10 billion in ten-year bonds with a 5.125% coupon, and the bids are as in table
- Bids will be filled from the lowest yield (highest price) until the entire \$10 billion has been raised.
- This auction will clear at a yield of 5.122 percent and all bidders will receive the same interest rate.
- Also used by online lending platforms such as [Prosper](#).

\$1.0 billion at 5.115%
\$2.5 billion at 5.117%
\$3.5 billion at 5.120%
\$4.5 billion at 5.122%
\$3.75 billion at 5.124%
\$3.75 billion at 5.130%

Summary

- Different strategies are set for different products, specially when the cost base differs
 - Fixed cost products require large items sales to recoup the sunk cost
 - Price discrimination methods try to capture as much of the market as possible
 - Variable cost products do not need to rely much on price discrimination – still they need to do that to capture as much profit as possible
- Different strategies are used to maximise revenue and profits